Final Project Design Decisions

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Scene Design and Object Construction

I chose the Mink Car cover by They Might Be Giants for my scene. The image shows a blue figure broken into parts, which suited a build from basic shapes. I picked each shape based on how close it looked to the part it represents. The head is a sphere, the torso and hands are boxes, and the arms and legs are cylinders. The neck and joints are tapered cylinders; the small bumps where parts connect are pyramids, and the feet are cones. The figure is made from several different shapes. Every piece uses the same paper texture, so the figure looks like a printed model.

The scene did not get as close to the image as I wanted. Rotating each piece and fitting it into its section of the figure was difficult because the image is flat and its curved outlines are hard to copy with basic shapes. Still, I think the result shows that I understand how to place, scale, rotate, and render objects in OpenGL. Next time, I would choose a 3D reference instead of a 2D one, since depth and angles would be easier to model.

Navigation and Camera Controls

The user moves around the scene with the keyboard and mouse. The W and S keys move the camera forward and backward, the A and D keys move it left and right, and the Q and E keys move it up and down. The mouse changes the camera direction. Moving it left or right turns the view, and moving it up or down tilts it, but the camera position stays the same. The scroll wheel changes the movement speed, which is kept between 1 and 30. Movement is multiplied by the time between frames, so the speed feels the same on any computer. The P key switches to the perspective view, and the O key switches to the orthographic view. The orthographic view keeps the camera position and looks straight ahead, which shows the figure flat like the original image. The Escape key closes the window.

Custom Functions and Code Organization

DefineObjectMaterials sets the diffuse color, specular color, and shininess for the figure and stores them under a tag. Any object can use that material by calling its tag, and a new material only needs a new entry. SetupSceneLights sets the position, ambient, diffuse, and specular values of each light and turns it on. I placed the main point light above the figure so light reaches all of the pieces. I added a second point light, gray with a slight blue tint, in front of the figure to light the side that faces the camera. Keeping the lighting in its own function means I can change or add lights without touching the drawing code. I also made small changes to the fragment shader so the textures showed correctly and the lighting worked with both the textures and the object colors.